Lecture 8

Sequencing Technologies.

FASTQ Quality Control.

Sequencing Technologies

1st generation: Frederic Sanger develops DNA sequencing technology. Latest versions of it generate 3 million bases/day, 1500bp long reads

2nd generation: (next-gen) sequencing started 2005 with the release of the 454 sequencing platform. 600 billion bases/week, 150bp long reads

3rd generation: single molecule (no DNA amplification required), these are not replacing but augmenting 2nd generation systems, longer reads, shorter turnarounds)

Field guide to next generation sequencing instrumentation

See lecture link to the **The Field Guide** by Travis Glenn. It answers questions of the type:

- How many kinds of instruments exists?
- How much data does each instrument produce?
- How much does it cost for reagents?
- How long does a sequencing run take?

There is a blog page with updates for 2016.

What is a "read"?

A **read** is a sequence measurement.

The sequencing instrument "reads" the DNA fragment.

A read has a size (length) and qualities associated with it.

We call sequences "reads" when in the context of measurements.

Some instruments produce reads of equal sizes even if the length of the DNA fragment varies.

Paired end sequencing

Instruments may produce more than one measurement for a single DNA fragment.

Current protocols for the Illumina instruments label this as "paired-end reads" and "mated-pair reads."

The PacBio tool may generate multiple 3-10 reads out of the same fragment.

Having two or more copies has many benefits but will add to the complexity of the analysis.

Later we will cover this in more detail.

Visualizing qualities

The FASTQ format contains qualities (probabilities).

Tools can visualize the distribution of these numbers.

The most commonly used tool `fastqc`

It is perhaps the most frequently used bioinformatics software.

Analyses should begin with a data quality evaluation.

Get a test dataset

Make your code "self" documenting.
Add comments after the `#` symbol.

```
# Obtain the test dataset.
wget http://data.biostarhandbook.com/data/sequencing-platf

# Uncompress the data.
tar xzvf sequencing-platform-data.tar.gz

# See what you have got.
ls
```

This archive contains data from different sequencing instruments. Files are named after the instrument.

Running FastQC

If you are on Windows you should run this in a directory that is visible in Windows. See the Windows section of the installation.

Run fastqc on the file called `illumina.fq` :

```
fastqc illumina.fq
```

prints some information then generates a file called:

```
illumina_fastqc.html
```

Open up this file in the web browser.

How to interpret the plots?

FastQC generates quite a few plots.

Some are easy to understand, others not so much.

There is a description for each analysis on the FastQC website.

The handbook also discusses many plots and expands on those where the official explanations were deemed insufficient.

A bit of self-study is required here.

FastQC: Per base sequence quality

FastQC Report

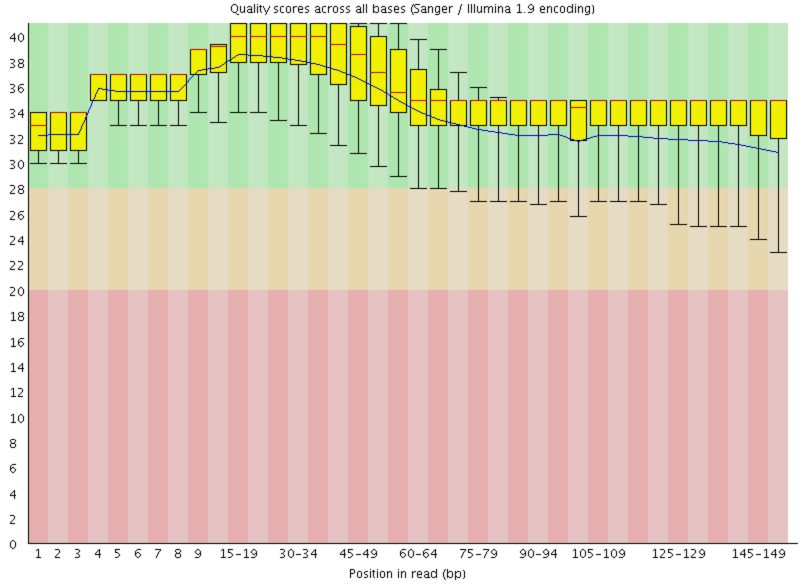
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content
- Kmer Content

Basic Statistics

Measure	Value
Filename	illumina.fq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	200000
Sequences flagged as poor quality	0
Sequence length	150
%GC	51

Per base sequence quality



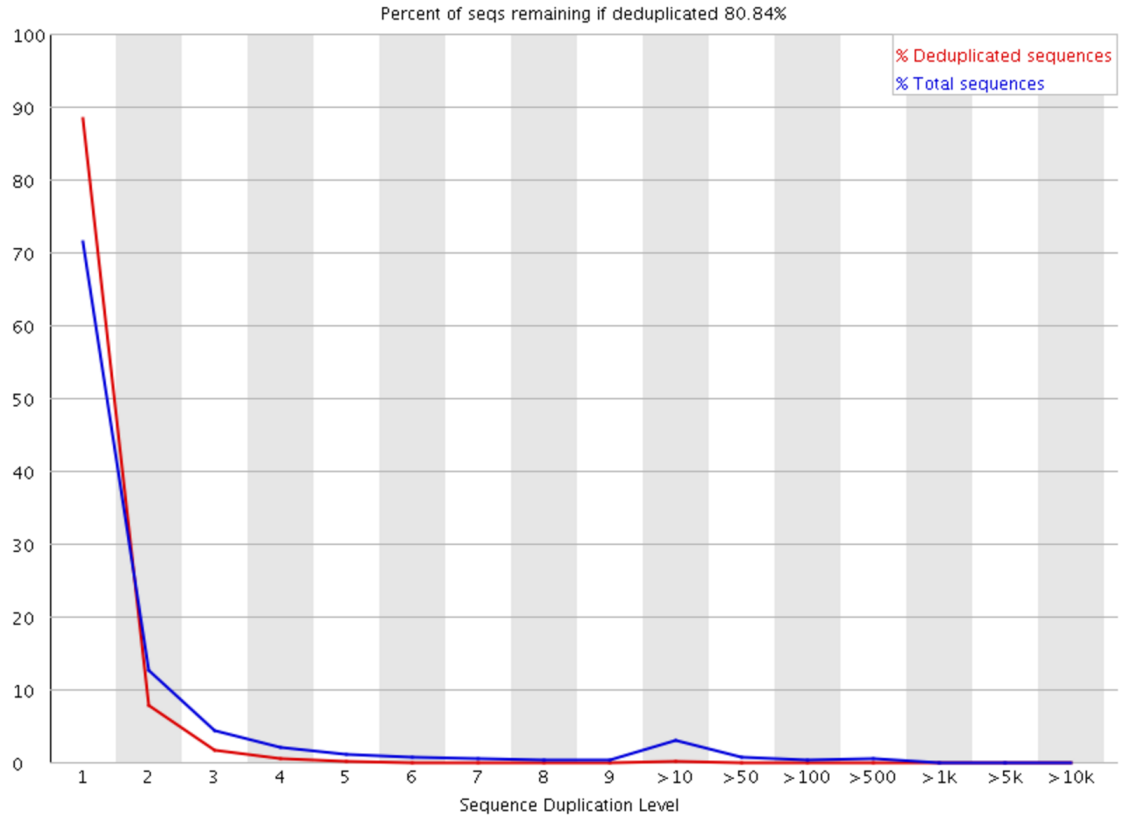
FastQC: Sequence duplication levels

FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
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- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content
- Kmer Content

Sequence Duplication Levels



FastQC Resources

Do consult the documentation and examples of good and bad data.

Documentation

A copy of the FastQC documentation is available for you to try before you buy (well download..).

Example Reports

- Good Illumina Data
- Bad Illumina Data
- Adapter dimer contaminated run
- Small RNA with read-through adapter
- Reduced Representation BS-Seq
- PacBio
- 454

Available on the `fastqc` website.

What is quality control?

Quality control (QC) is the process of improving data by removing identifiable errors from it.

It is typically the first step performed after data acquisition.

Note: QC will alter data and may introduce biases and artifacts! There is a tradeoff.

Apply QC only when you have problems that need fixing!

How do we perform quality control?

QC typically follows this process.

1. Evaluate (visualize) data quality.
2. Stop QC if the quality appears to be satisfactory.
3. If the quality is still inadequate, execute one or more data altering steps then go to step 1.

There are different software that you can use.

We like `trimmomatic`, `cutadapt`, `bbduk`

Quality control steps

In some instruments, the quality deteriorates towards the end. Common quality control steps:

1. Keep reads with an average quality over a threshold.
2. Trim back the ends of reads until the measurements are reliable.
3. Remove known synthetic sequences.

If your data is paired-end, you have process both pairs and ensure to keep them in sync.

Trimming back reads by quality

Different strategies exist. If you start with:

```
sequence: ATGCATTTATATAT
quality:  II$&III!@$#II!
```

One trimming strategy coudl produce:

```
sequence: ATGCATTTATATA
quality:  II$&III!@$#II
```

Another trimming strategy might give you:

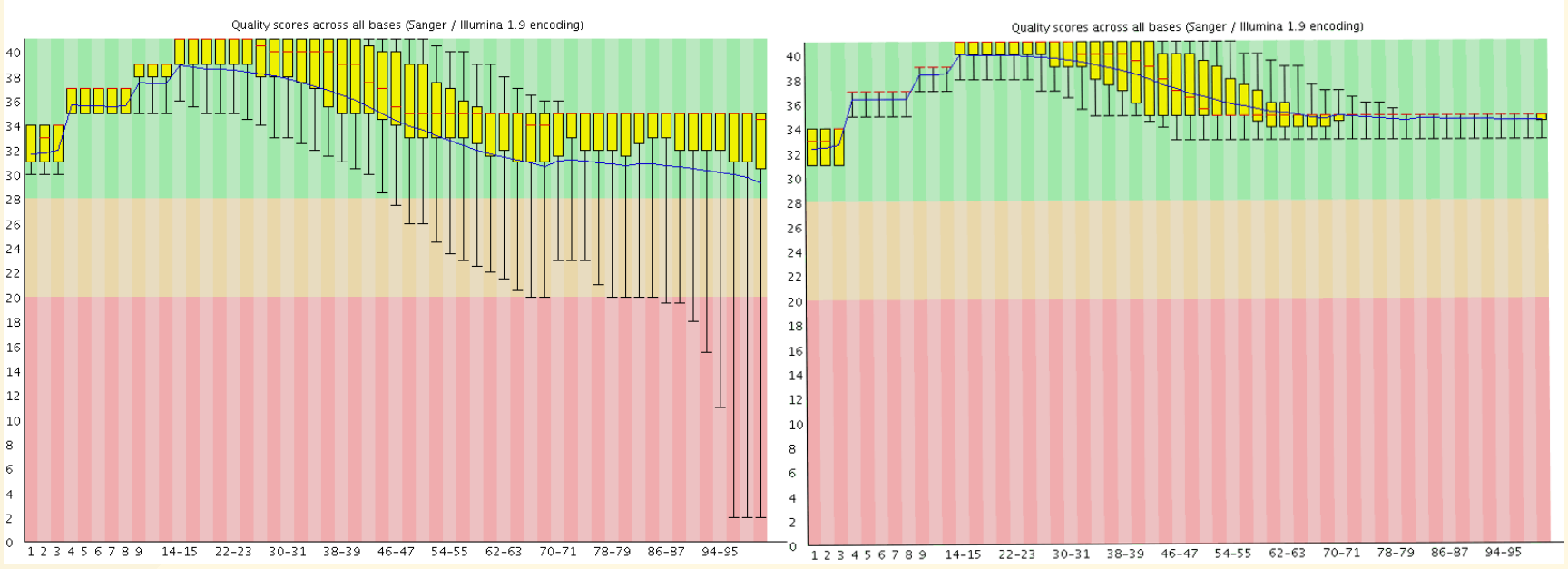
```
sequence: ATGCATT
quality:  II$&III
```

How do I run a quality trimming?

Trimmomatic sliding window trimming:

```
trimmomatic SE illumina.fq better_data.fq
                                    SLIDINGWINDOW:4:30
```

Compare the results:



Adapter contamination

An **adapter**: a short, known, usually synthetic sequence. Usually added during the lab prep.

For the instrument to work there has to be known information in the sequences, usually at each end.

Instruments read off a certain length of DNA from the fragment.

Reads that are longer than the original fragment will run into the artificial sequences.

Other reasons for having adapters

Biologists are getting very good at marking DNA at various locations.

Numerous techniques operate by inserting known sequences into DNA.

The same adapter identification techniques may be used there as well.

Suppose that the adapter is `ATGATGATGATG` then a read may look like:

```
ATGCATTTATATATATGATGATG
real sequence | adapter
```

Removing adapters

Keep original data, Remove adapter `ATGATGATGATG`

```
ATGCATTTATATATATGATGATG
real sequence | adapter
```

to produce:

```
ATGCATTTATATAT
real sequence
```

The overlap has to be long enough

We have to be able to recognize the adapter reliably.

The sequence below has an adapter, but we would not know for sure since we only have two bases from it:

```
ATGCATTTATATATAT
real sequence | adapter
```

The adapter should also be unique to the situation.

Someone once messed up a dataset by attempting to remove polyadenylation (runs of `AAAA`). Guess what runs of `AAAA` are also legitimate DNA.

How do I remove adapters?

You need an adapter sequence file. See the book for code that you can copy paste:

```
echo ">illumina" > adapter.fa
echo "AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC" >> adapter.fa
```

Trimming adapters with `trimmomatic`:

```
trimmomatic SE illumina.fq better_data.fq
                          ILLUMINACLIP:adapter.fa:2:30:5
```

Choices, choices.

You can apply multiple actions as well:

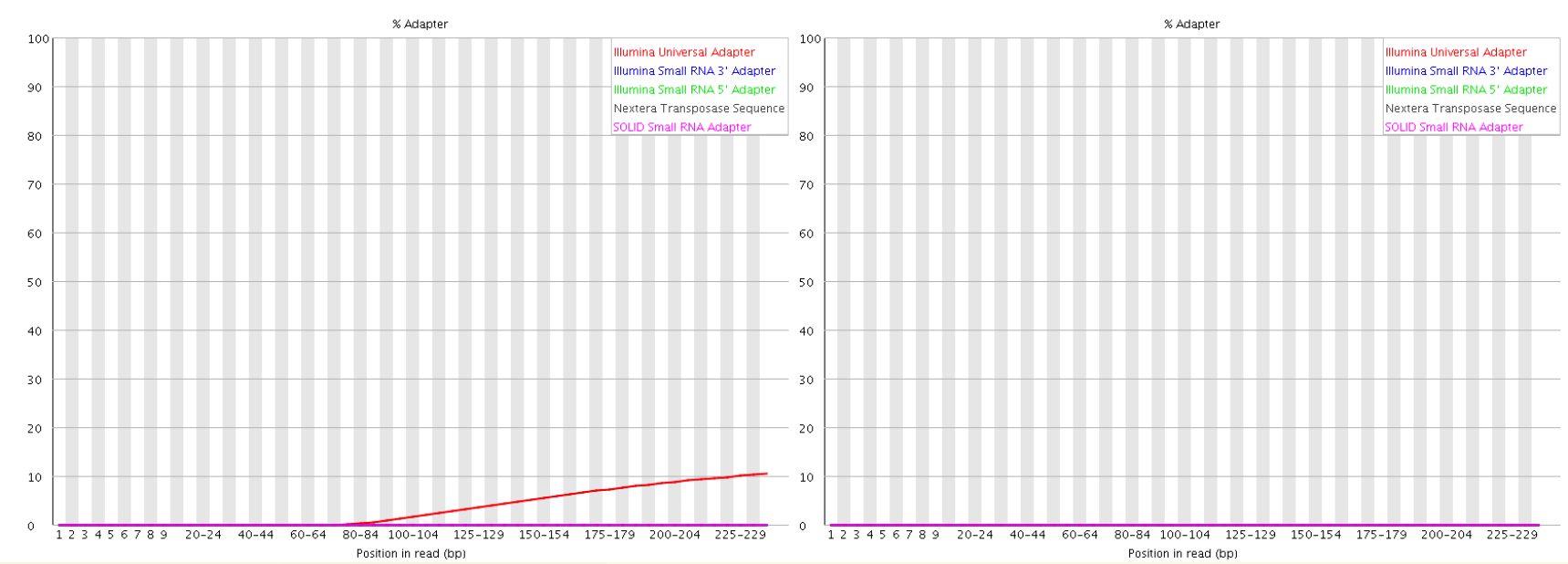
```
trimmomatic SE illumina.fq better_data.fq
       ILLUMINACLIP:adapter.fa:2:30:5 SLIDINGWINDOW:4:30
```

Does matter which action comes first? What should come first? Trim adapters then trim for qualities or vice versa?

You can also another tool:

```
bbduk.sh in1=illumina.fq  out=better_data.fq qtrim=30
                                              overwrite=true
```

Effect of adapter trimming



`fastqc` knows about certain types of adapters.

The adapter sequence detection can be customized in `fastqc`.

Important to remember!

QC alters data! It is not without risk.

QC moves data toward a preconceived notion of what real data "ought" to be like.

Apply QC if your data needs it. Evaluate results before and after.

Adapter trimming, if done incorrectly, can destroy the meaning of your data!

Duplicate removal, if done incorrectly, can radically change your results!